

МИНОБРНАУКИ РОССИИ
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«МИРЭА – Российский технологический университет»

РТУ МИРЭА

Институт информационных технологий (ИИТ)
Кафедра цифровой трансформации (ЦТ)

ОТЧЕТ ПО ПРАКТИЧЕСКОЙ РАБОТЕ

по дисциплине «Проектирование баз данных»

Практическая работа № 6

Студент группы ИВБО-20-24 Лияскин Владислав Андреевич _____
(подпись)

Ассистент Шерри Георгий Даниэлевич

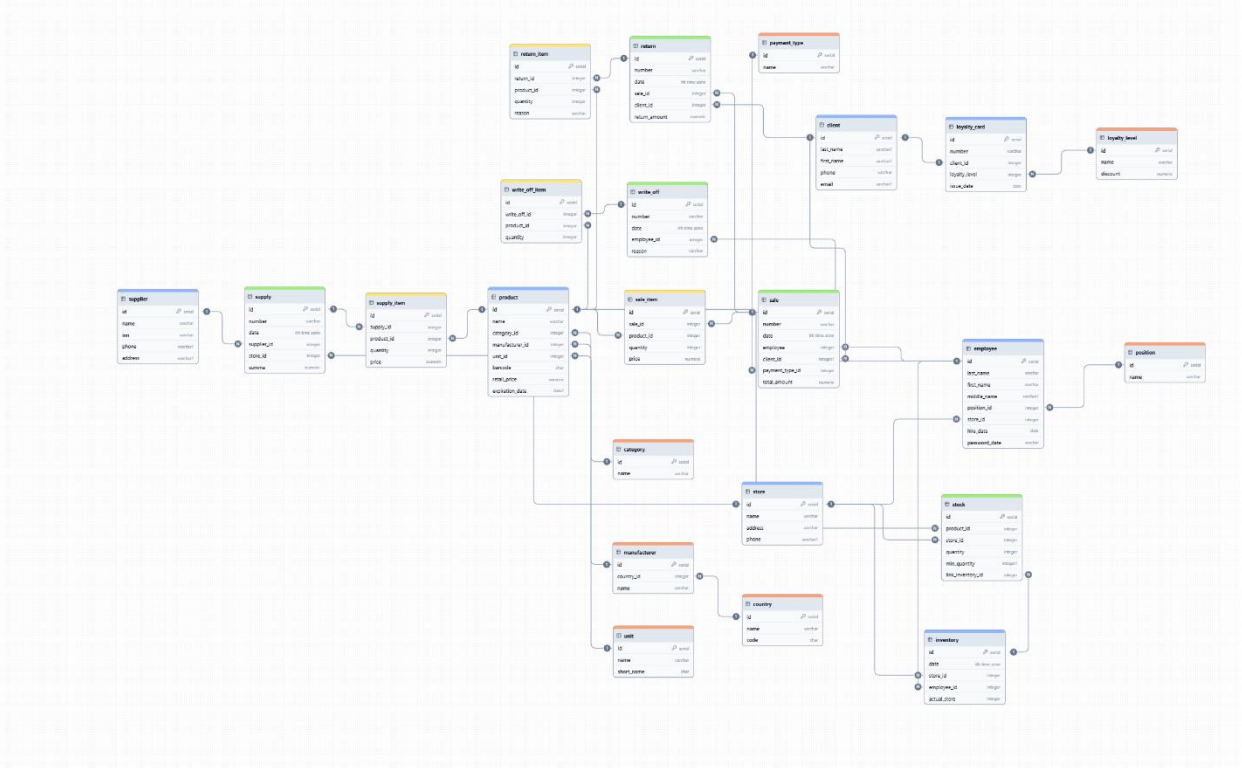
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Задание №1

Готовая диаграмма, выполненная по методичке



Задание №2

Вот код, который мне выдал ChartDB

```
CREATE TABLE write_off (
  id integer NOT NULL PRIMARY KEY,
  number varchar(500) NOT NULL UNIQUE,
  date timestamp with time zone NOT NULL,
  employee_id integer NOT NULL,
  reason varchar(500) NOT NULL
);


CREATE TABLE paymant_type (
  id integer NOT NULL PRIMARY KEY,
  name varchar(500) NOT NULL UNIQUE
);


CREATE TABLE stock (
  id serial NOT NULL PRIMARY KEY,
  product_id integer NOT NULL,
  store_id integer NOT NULL,
  quantity integer NOT NULL,
  min_quantity integer,
```

```
  link_inventory_id integer NOT NULL
);


CREATE TABLE sale (
  id integer NOT NULL PRIMARY KEY,
  number varchar(500) NOT NULL UNIQUE,
  date timestamp with time zone NOT NULL,
  employee integer NOT NULL,
  client_id integer,
  payment_type_id integer NOT NULL,
  total_amount numeric NOT NULL
);


CREATE TABLE sale_item (
  id serial NOT NULL PRIMARY KEY,
  sale_id integer NOT NULL,
  product_id integer NOT NULL,
  quantity integer NOT NULL,
  price numeric NOT NULL
);


CREATE TABLE product (
  id integer NOT NULL PRIMARY KEY,
  name varchar(500) NOT NULL,
  category_id integer NOT NULL,
  manufacturer_id integer NOT NULL,
  unit_id integer NOT NULL,
  barcode char NOT NULL UNIQUE,
  retail_price numeric NOT NULL,
  expiration_date date
);


CREATE TABLE supplier (
  id integer NOT NULL PRIMARY KEY,
  name varchar(500) NOT NULL UNIQUE,
  inn varchar(500) NOT NULL UNIQUE,
  phone varchar(500),
  address varchar(500)
);


CREATE TABLE loyalty_level (
  id integer NOT NULL PRIMARY KEY,
  name varchar(500) NOT NULL UNIQUE,
  discount numeric NOT NULL
);


CREATE TABLE loyalty_card (
  id serial NOT NULL PRIMARY KEY,
  number varchar(500) NOT NULL UNIQUE,
  client_id integer NOT NULL,
  loyalty_level integer NOT NULL,
  issue_date date NOT NULL
);


CREATE TABLE return_item (
  id serial NOT NULL PRIMARY KEY,
  return_id integer NOT NULL,
  product_id integer NOT NULL,
  quantity integer NOT NULL,
  reason varchar(500) NOT NULL
);
```

```
CREATE TABLE store (
  id integer NOT NULL PRIMARY KEY,
  name varchar(500) NOT NULL UNIQUE,
  address varchar(500) NOT NULL,
  phone varchar(500)
);


CREATE TABLE write_off_item (
  id serial NOT NULL PRIMARY KEY,
  write_off_id integer NOT NULL,
  product_id integer NOT NULL,
  quantity integer NOT NULL
);


CREATE TABLE manufacturer (
  id integer NOT NULL PRIMARY KEY,
  country_id integer NOT NULL,
  name varchar(500) NOT NULL UNIQUE
);


CREATE TABLE supply_item (
  id serial NOT NULL PRIMARY KEY,
  supply_id integer NOT NULL,
  product_id integer NOT NULL,
  quantity integer NOT NULL,
  price numeric NOT NULL
);


CREATE TABLE category (
  id integer NOT NULL PRIMARY KEY,
  name varchar(500) NOT NULL UNIQUE
);


CREATE TABLE country (
  id integer NOT NULL PRIMARY KEY,
  name varchar(500) NOT NULL UNIQUE,
  code char NOT NULL UNIQUE
);


CREATE TABLE supply (
  id integer NOT NULL PRIMARY KEY,
  number varchar(500) NOT NULL UNIQUE,
  date timestamp with time zone NOT NULL,
  supplier_id integer NOT NULL,
  store_id integer NOT NULL,
  summa numeric NOT NULL
);


CREATE TABLE client (
  id integer NOT NULL PRIMARY KEY,
  last_name varchar(500),
  first_name varchar(500),
  phone varchar(500) NOT NULL UNIQUE,
  email varchar(500) UNIQUE
);


CREATE TABLE position (
  id integer NOT NULL PRIMARY KEY,
  name varchar(500) NOT NULL UNIQUE
);
```

```
CREATE TABLE unit (
  id integer NOT NULL PRIMARY KEY,
  name varchar(500) NOT NULL UNIQUE,
  short_name char NOT NULL UNIQUE
);


CREATE TABLE return (
  id integer NOT NULL PRIMARY KEY,
  number varchar(500) NOT NULL UNIQUE,
  date timestamp with time zone NOT NULL,
  sale_id integer NOT NULL,
  client_id integer NOT NULL,
  return_amount numeric NOT NULL
);


CREATE TABLE employee (
  id integer NOT NULL PRIMARY KEY,
  last_name varchar(500) NOT NULL,
  first_name varchar(500) NOT NULL,
  middle_name varchar(500),
  position_id integer NOT NULL,
  store_id integer NOT NULL,
  hire_date date NOT NULL,
  password_date varchar(500) NOT NULL UNIQUE
);


CREATE TABLE inventory (
  id integer NOT NULL PRIMARY KEY,
  date timestamp with time zone NOT NULL,
  store_id integer NOT NULL,
  employee_id integer NOT NULL,
  actual_store integer NOT NULL
);


ALTER TABLE supply ADD CONSTRAINT supply_supplier_id_fk FOREIGN KEY (supplier_id) REFERENCES supplier
(id);
ALTER TABLE supply_item ADD CONSTRAINT supply_item_supply_id_fk FOREIGN KEY (supply_id) REFERENCES
supply (id);
ALTER TABLE product ADD CONSTRAINT product_category_id_fk FOREIGN KEY (category_id) REFERENCES
category (id);
ALTER TABLE product ADD CONSTRAINT product_manufacturer_id_fk FOREIGN KEY (manufacturer_id)
REFERENCES manufacturer (id);
ALTER TABLE product ADD CONSTRAINT product_unit_id_fk FOREIGN KEY (unit_id) REFERENCES unit (id);
ALTER TABLE manufacturer ADD CONSTRAINT manufacturer_country_id_fk FOREIGN KEY (country_id)
REFERENCES country (id);
ALTER TABLE employee ADD CONSTRAINT employee_position_id_fk FOREIGN KEY (position_id) REFERENCES
position (id);
ALTER TABLE employee ADD CONSTRAINT employee_store_id_fk FOREIGN KEY (store_id) REFERENCES store
(id);
ALTER TABLE loyalty_card ADD CONSTRAINT loyalty_card_client_id_fk FOREIGN KEY (client_id) REFERENCES
client (id);
ALTER TABLE loyalty_card ADD CONSTRAINT loyalty_card_loyalty_level_fk FOREIGN KEY (loyalty_level)
REFERENCES loyalty_level (id);
ALTER TABLE supply_item ADD CONSTRAINT supply_item_product_id_fk FOREIGN KEY (product_id)
REFERENCES product (id);
ALTER TABLE sale ADD CONSTRAINT sale_employee_fk FOREIGN KEY (employee) REFERENCES employee (id);
ALTER TABLE sale ADD CONSTRAINT sale_client_id_fk FOREIGN KEY (client_id) REFERENCES client (id);
ALTER TABLE sale ADD CONSTRAINT sale_payment_type_id_fk FOREIGN KEY (payment_type_id) REFERENCES
paymant_type (id);
ALTER TABLE sale_item ADD CONSTRAINT sale_item_sale_id_fk FOREIGN KEY (sale_id) REFERENCES sale (id);
ALTER TABLE sale_item ADD CONSTRAINT sale_item_product_id_fk FOREIGN KEY (product_id) REFERENCES
product (id);
ALTER TABLE write_off ADD CONSTRAINT write_off_employee_id_fk FOREIGN KEY (employee_id) REFERENCES
employee (id);
ALTER TABLE return ADD CONSTRAINT return_sale_id_fk FOREIGN KEY (sale_id) REFERENCES sale (id);
ALTER TABLE return ADD CONSTRAINT return_client_id_fk FOREIGN KEY (client_id) REFERENCES client (id);
```

```
ALTER TABLE supply ADD CONSTRAINT supply_store_id_fk FOREIGN KEY (store_id) REFERENCES store (id);
ALTER TABLE write_off_item ADD CONSTRAINT write_off_item_write_off_id_fk FOREIGN KEY (write_off_id) REFERENCES write_off (id);
ALTER TABLE write_off_item ADD CONSTRAINT write_off_item_product_id_fk FOREIGN KEY (product_id) REFERENCES product (id);
ALTER TABLE return_item ADD CONSTRAINT return_item_return_id_fk FOREIGN KEY (return_id) REFERENCES return (id);
ALTER TABLE return_item ADD CONSTRAINT return_item_product_id_fk FOREIGN KEY (product_id) REFERENCES product (id);
ALTER TABLE stock ADD CONSTRAINT stock_product_id_fk FOREIGN KEY (product_id) REFERENCES product (id);
ALTER TABLE stock ADD CONSTRAINT stock_store_id_fk FOREIGN KEY (store_id) REFERENCES store (id);
ALTER TABLE stock ADD CONSTRAINT stock_link_inventory_id_fk FOREIGN KEY (link_inventory_id) REFERENCES inventory (id);
ALTER TABLE inventory ADD CONSTRAINT inventory_store_id_fk FOREIGN KEY (store_id) REFERENCES store (id);
ALTER TABLE inventory ADD CONSTRAINT inventory_employee_id_fk FOREIGN KEY (employee_id) REFERENCES employee (id);
```

Задание №3

Вот доработанный код из задания №2

```
-УРОВЕНЬ 1: НЕЗАВИСИМЫЕ СПРАВОЧНИКИ (без внешних ключей)

CREATE TABLE IF NOT EXISTS country (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(50) NOT NULL UNIQUE,
    code char(2) NOT NULL UNIQUE
);

CREATE TABLE IF NOT EXISTS unit (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(30) NOT NULL UNIQUE,
    short_name char(5) NOT NULL UNIQUE
);

CREATE TABLE IF NOT EXISTS category (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(50) NOT NULL UNIQUE
);

CREATE TABLE IF NOT EXISTS payment_type (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(30) NOT NULL UNIQUE
);

CREATE TABLE IF NOT EXISTS position (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(50) NOT NULL UNIQUE
);

CREATE TABLE IF NOT EXISTS loyalty_level (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(30) NOT NULL UNIQUE,
    discount numeric(5, 2) NOT NULL DEFAULT 0.00
);
COMMENT ON TABLE loyalty_level IS 'Справочник уровней лояльности с процентом скидки';


-- УРОВЕНЬ 2: ЗАВИСИМЫЕ СПРАВОЧНИКИ И БАЗОВЫЕ ОБЪЕКТЫ

CREATE TABLE IF NOT EXISTS manufacturer (
```

```
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(100) NOT NULL UNIQUE,
    country_id int NOT NULL
);

CREATE TABLE IF NOT EXISTS store (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(100) NOT NULL UNIQUE,
    address varchar(200) NOT NULL,
    phone varchar(20)
);

CREATE TABLE IF NOT EXISTS supplier (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(100) NOT NULL UNIQUE,
    inn varchar(12) NOT NULL UNIQUE,
    phone varchar(20),
    address varchar(200)
);

CREATE TABLE IF NOT EXISTS client (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    last_name varchar(50),
    first_name varchar(50),
    phone varchar(20) NOT NULL UNIQUE,
    email varchar(255) UNIQUE
);


-- УРОВЕНЬ 3: КЛЮЧЕВЫЕ ОБЪЕКТЫ (зависят от справочников)

CREATE TABLE IF NOT EXISTS product (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name varchar(100) NOT NULL,
    category_id int NOT NULL,
    manufacturer_id int NOT NULL,
    unit_id int NOT NULL,
    barcode char(13) NOT NULL UNIQUE,
    retail_price numeric(10, 2) NOT NULL DEFAULT 0.00,
    expiration_date date
);

CREATE TABLE IF NOT EXISTS employee (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    last_name varchar(50) NOT NULL,
    first_name varchar(50) NOT NULL,
    middle_name varchar(50),
    position_id int NOT NULL,
    store_id int NOT NULL,
    hire_date date NOT NULL DEFAULT CURRENT_DATE,
    password varchar(255) NOT NULL UNIQUE
);

CREATE TABLE IF NOT EXISTS loyalty_card (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    number varchar(16) NOT NULL UNIQUE,
    client_id int NOT NULL,
    level_id int NOT NULL,
    issue_date date NOT NULL DEFAULT CURRENT_DATE
);


-- УРОВЕНЬ 4: ДОКУМЕНТЫ И РЕГИСТРЫ (шапки бизнес-событий)

CREATE TABLE IF NOT EXISTS inventory (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    date timestamptz NOT NULL DEFAULT CURRENT_TIMESTAMP,
    store_id int NOT NULL,
    employee_id int NOT NULL,
    actual_stock int NOT NULL DEFAULT 0
```

```
);

CREATE TABLE IF NOT EXISTS stock (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    product_id int NOT NULL,
    store_id int NOT NULL,
    quantity int NOT NULL DEFAULT 0,
    min_quantity int DEFAULT 0,
    link_inventory_id int
);
COMMENT ON TABLE stock IS 'Регистр текущих складских остатков по филиалам';

CREATE TABLE IF NOT EXISTS supply (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    number varchar(20) NOT NULL UNIQUE,
    date timestamptz NOT NULL DEFAULT CURRENT_TIMESTAMP,
    supplier_id int NOT NULL,
    store_id int NOT NULL,
    summa numeric(10, 2) NOT NULL DEFAULT 0.00
);

CREATE TABLE IF NOT EXISTS sale (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    number varchar(20) NOT NULL UNIQUE,
    date timestamptz NOT NULL DEFAULT CURRENT_TIMESTAMP,
    employee_id int NOT NULL,
    client_id int,
    payment_type_id int NOT NULL,
    total_amount numeric(10, 2) NOT NULL DEFAULT 0.00
);
COMMENT ON TABLE sale IS 'Шапка документа кассовой продажи (чека)';

CREATE TABLE IF NOT EXISTS write_off (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    number varchar(20) NOT NULL UNIQUE,
    date timestamptz NOT NULL DEFAULT CURRENT_TIMESTAMP,
    employee_id int NOT NULL,
    reason varchar(200) NOT NULL
);

CREATE TABLE IF NOT EXISTS return (
    id int GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    number varchar(20) NOT NULL UNIQUE,
    date timestamptz NOT NULL DEFAULT CURRENT_TIMESTAMP,
    sale_id int NOT NULL,
    client_id int NOT NULL,
    return_amount numeric(10, 2) NOT NULL DEFAULT 0.00
);


-- УРОВЕНЬ 5: ПРОМЕЖУТОЧНЫЕ ТАБЛИЦЫ (позиции документов, слабые сущности)

CREATE TABLE IF NOT EXISTS supply_item (
    supply_id int NOT NULL,
    product_id int NOT NULL,
    quantity int NOT NULL DEFAULT 1,
    price numeric(10, 2) NOT NULL DEFAULT 0.00,
    PRIMARY KEY (supply_id, product_id)
);
COMMENT ON TABLE supply_item IS 'Позиции документа поставки';

CREATE TABLE IF NOT EXISTS sale_item (
    sale_id int NOT NULL,
    product_id int NOT NULL,
    quantity int NOT NULL DEFAULT 1,
    price numeric(10, 2) NOT NULL DEFAULT 0.00,
    PRIMARY KEY (sale_id, product_id)
);

CREATE TABLE IF NOT EXISTS write_off_item (
```

```
    write_off_id int NOT NULL,
    product_id int NOT NULL,
    quantity int NOT NULL DEFAULT 1,
    PRIMARY KEY (write_off_id, product_id)
);

CREATE TABLE IF NOT EXISTS return_item (
    return_id int NOT NULL,
    product_id int NOT NULL,
    quantity int NOT NULL DEFAULT 1,
    reason varchar(200) NOT NULL,
    PRIMARY KEY (return_id, product_id)
);


-- БЛОК ОГРАНИЧЕНИЙ ВНЕШНИХ КЛЮЧЕЙ (ON DELETE / ON UPDATE)


-- 1. Защита справочников и ключевых объектов (ON DELETE RESTRICT)
ALTER TABLE manufacturer ADD CONSTRAINT fk_manufacturer_country FOREIGN KEY (country_id) REFERENCES
country (id) ON DELETE RESTRICT;
ALTER TABLE product ADD CONSTRAINT fk_product_category FOREIGN KEY (category_id) REFERENCES category
(id) ON DELETE RESTRICT;
ALTER TABLE product ADD CONSTRAINT fk_product_manufacturer FOREIGN KEY (manufacturer_id) REFERENCES
manufacturer (id) ON DELETE RESTRICT;
ALTER TABLE product ADD CONSTRAINT fk_product_unit FOREIGN KEY (unit_id) REFERENCES unit (id) ON
DELETE RESTRICT;
ALTER TABLE employee ADD CONSTRAINT fk_employee_position FOREIGN KEY (position_id) REFERENCES
position (id) ON DELETE RESTRICT;
ALTER TABLE employee ADD CONSTRAINT fk_employee_store FOREIGN KEY (store_id) REFERENCES store (id)
ON DELETE RESTRICT;
ALTER TABLE loyalty_card ADD CONSTRAINT fk_loyalty_card_client FOREIGN KEY (client_id) REFERENCES client
(id) ON DELETE RESTRICT;
ALTER TABLE loyalty_card ADD CONSTRAINT fk_loyalty_card_level FOREIGN KEY (level_id) REFERENCES
loyalty_level (id) ON DELETE RESTRICT;
ALTER TABLE inventory ADD CONSTRAINT fk_inventory_store FOREIGN KEY (store_id) REFERENCES store (id)
ON DELETE RESTRICT;
ALTER TABLE inventory ADD CONSTRAINT fk_inventory_employee FOREIGN KEY (employee_id) REFERENCES
employee (id) ON DELETE RESTRICT;
ALTER TABLE stock ADD CONSTRAINT fk_stock_product FOREIGN KEY (product_id) REFERENCES product (id) ON
DELETE RESTRICT;
ALTER TABLE stock ADD CONSTRAINT fk_stock_store FOREIGN KEY (store_id) REFERENCES store (id) ON
DELETE RESTRICT;
ALTER TABLE stock ADD CONSTRAINT fk_stock_inventory FOREIGN KEY (link_inventory_id) REFERENCES
inventory (id) ON DELETE RESTRICT;
ALTER TABLE supply ADD CONSTRAINT fk_supply_supplier FOREIGN KEY (supplier_id) REFERENCES supplier (id)
ON DELETE RESTRICT;
ALTER TABLE supply ADD CONSTRAINT fk_supply_store FOREIGN KEY (store_id) REFERENCES store (id) ON
DELETE RESTRICT;
ALTER TABLE sale ADD CONSTRAINT fk_sale_employee FOREIGN KEY (employee_id) REFERENCES employee (id)
ON DELETE RESTRICT;
ALTER TABLE sale ADD CONSTRAINT fk_sale_client FOREIGN KEY (client_id) REFERENCES client (id) ON
DELETE RESTRICT;
ALTER TABLE sale ADD CONSTRAINT fk_sale_payment FOREIGN KEY (payment_type_id) REFERENCES
payment_type (id) ON DELETE RESTRICT;
ALTER TABLE write_off ADD CONSTRAINT fk_write_off_employee FOREIGN KEY (employee_id) REFERENCES
employee (id) ON DELETE RESTRICT;
ALTER TABLE return ADD CONSTRAINT fk_return_sale FOREIGN KEY (sale_id) REFERENCES sale (id) ON
DELETE RESTRICT;
ALTER TABLE return ADD CONSTRAINT fk_return_client FOREIGN KEY (client_id) REFERENCES client (id) ON
DELETE RESTRICT;

-- 2. Целостность документов: позиции удаляются каскадно (ON DELETE CASCADE)
-- Для демонстрации навыка добавлен ON UPDATE CASCADE к supply_item и sale_item
ALTER TABLE supply_item ADD CONSTRAINT fk_supply_item_supply FOREIGN KEY (supply_id) REFERENCES
supply (id) ON DELETE CASCADE ON UPDATE CASCADE;
ALTER TABLE supply_item ADD CONSTRAINT fk_supply_item_product FOREIGN KEY (product_id) REFERENCES
product (id) ON DELETE RESTRICT;

ALTER TABLE sale_item ADD CONSTRAINT fk_sale_item_sale FOREIGN KEY (sale_id) REFERENCES sale (id) ON
```

```
DELETE CASCADE ON UPDATE CASCADE;
ALTER TABLE sale_item ADD CONSTRAINT fk_sale_item_product FOREIGN KEY (product_id) REFERENCES product (id) ON DELETE RESTRICT;

ALTER TABLE write_off_item ADD CONSTRAINT fk_write_off_item_write_off FOREIGN KEY (write_off_id) REFERENCES write_off (id) ON DELETE CASCADE;
ALTER TABLE write_off_item ADD CONSTRAINT fk_write_off_item_product FOREIGN KEY (product_id) REFERENCES product (id) ON DELETE RESTRICT;

ALTER TABLE return_item ADD CONSTRAINT fk_return_item_return FOREIGN KEY (return_id) REFERENCES return (id) ON DELETE CASCADE;
ALTER TABLE return_item ADD CONSTRAINT fk_return_item_product FOREIGN KEY (product_id) REFERENCES product (id) ON DELETE RESTRICT;
```

Заключение

В ходе выполнения практической работы №6 был осуществлён полный и корректный переход от логической реляционной модели данных к оптимизированному физическому DDL-скрипту для СУБД PostgreSQL. На основе инфологической модели, разработанной в Практической работе №2 для предметной области «Магазин напитков», с использованием CASE-средства ChartDB была визуализирована структура базы данных, где абстрактные атрибуты преобразованы в строгие физические типы данных с учётом бизнес-ограничений и прогнозируемых объёмов хранимой информации.